

Arrays in Java

Lecture 5

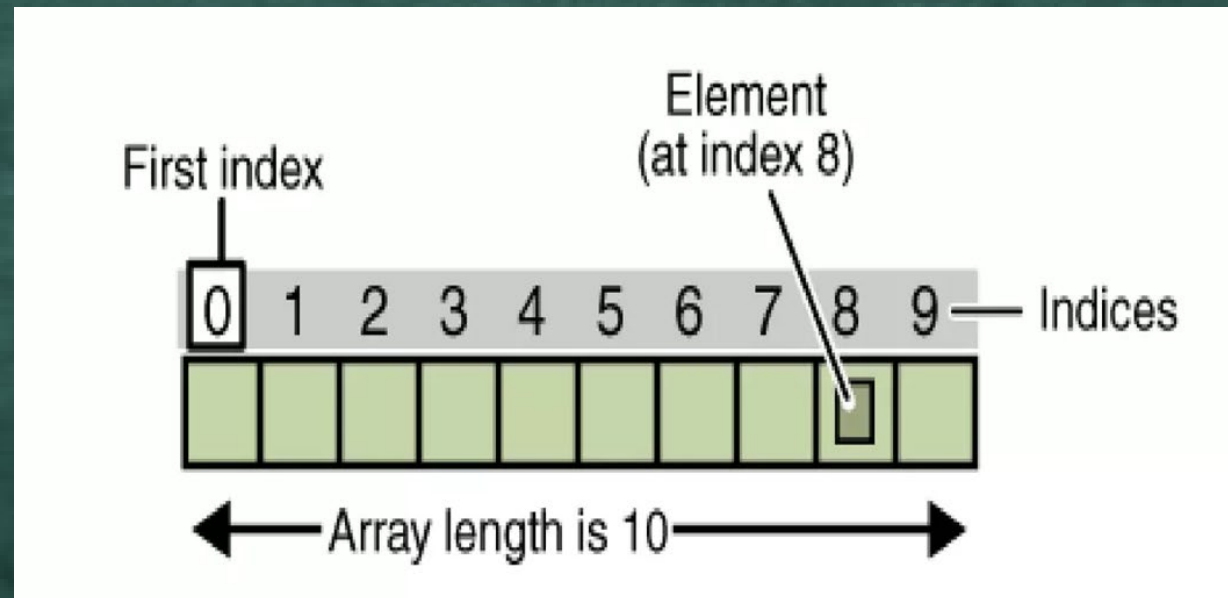
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Array

- Array is used to store a collection of variables of the same type.
- Memory Space is allocated for values of a particular type by declaring an array.
- In Java the array is index based started with 0. Must be non-negative. Integer.



Array Advantages and Disadvantages

Advantages

- Code Optimization.
- Random Access.

Disadvantages

- Size Limit.

Types of Array

- There are two types of array:
 - Single/One dimensional Array
 - Multidimensional Array

1 D ARRAY:

C	O	D	I	N	G	E	E	K
0	1	2	3	4	5	6	7	8

← single row of elements

2 D ARRAY:

		col 0	col 1	col 2	
	i \ j	0	1	2	← column
row 0	0	A	A	A	} array elements
row 1	1	B	B	B	
row 2	2	C	C	C	

↑
ROWS

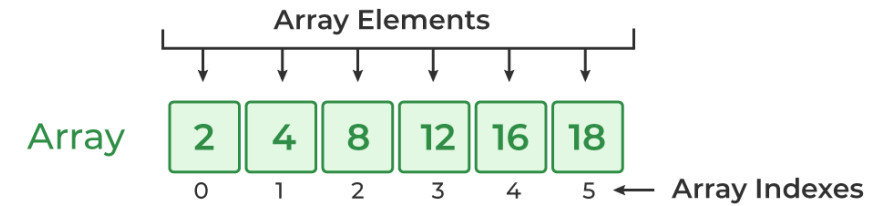
One Dimensional Array in C

- A single Array that holds multiple value of the same data type.

- Syntax to Declare a 1D Array:

- `datatype[] arrayName;`
- `datatype []arrayName;`
- `datatype arrayName[];`

One Dimensional Array in C



Creation of 1D Array

- `dataType [] arrayName = new dataType [arraySize];`

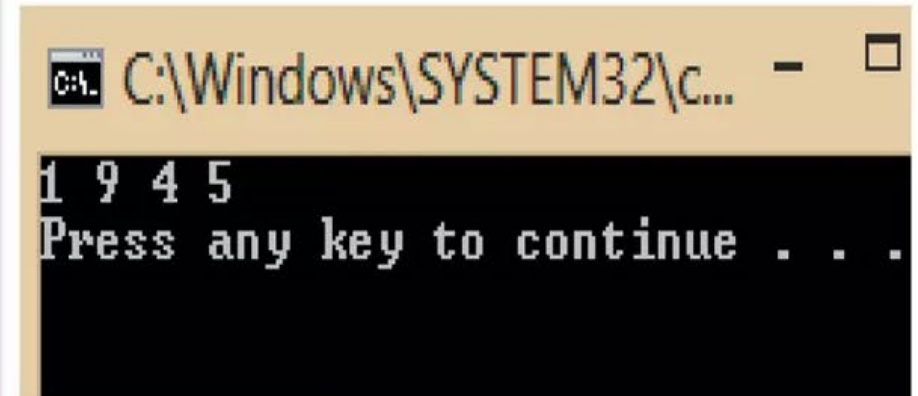
Or,

- `dataType[] arrayName={value0, value1, value2,.....};`

```
double[] a;           // declare the array
a = new double[n];     // create the array
for (int i = 0; i < n; i++) // elements are indexed from 0 to n-1
    a[i] = 0.0;        // initialize all elements to 0.0
```

Arrays and Loops

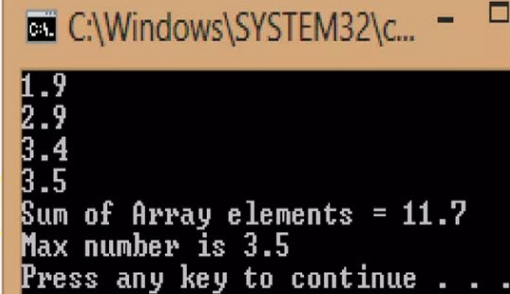
```
public class TestArray {  
    public static void main(String[] args) {  
  
        int[] myList = {1, 9, 4, 5};  
        for (int i = 0; i < myList.length; i++) {  
            System.out.print(myList[i] + " ");  
            System.out.println();  
        }  
    }  
}
```



A screenshot of a Windows command prompt window. The title bar shows the path 'C:\Windows\SYSTEM32\c...'. The command prompt displays the output of the Java program: '1 9 4 5' on the first line and 'Press any key to continue . . .' on the second line.

Array-Print, Sum, Max

```
public class Array_Sum_Max {  
    public static void main(String[] args) {  
        double[] myList = {1.9, 2.9, 3.4, 3.5};  
        // Print all the array elements  
        for(int i = 0; i < myList.length; i++){  
            System.out.println(myList[i] + " ");  
        }  
        // Summing all elements  
        double sum = 0;  
        for(int i = 0; i < myList.length; i++){  
            sum += myList[i];  
        }  
        System.out.println("Sum of Array elements = " + sum);  
        // Finding the largest element  
        double max = myList[0];  
        for (int i = 1; i < myList.length; i++) {  
            if (myList[i] > max) max = myList[i];  
        }  
        System.out.println("Max number is " + max);  
    }  
}
```



C:\Windows\SYSTEM32\c... - □

```
1.9  
2.9  
3.4  
3.5  
Sum of Array elements = 11.7  
Max number is 3.5  
Press any key to continue . . .
```

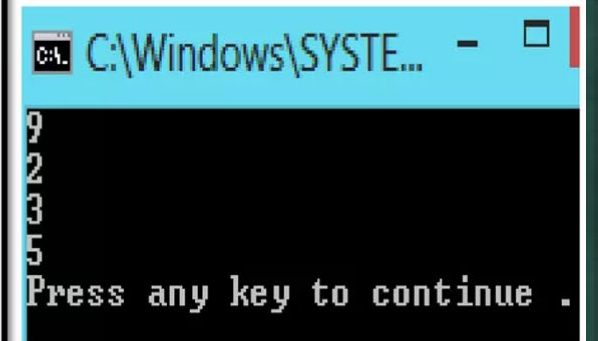

foreach loop in Array

- A new for loop known as foreach loop, enables you to traverse the complete array sequentially without using an index variable.
- **Syntax:**
- **Example:**

```
for (type var : array)
{
    statements using var;
}
```

```
public class ForEach_Array {
    public static void main(String[] args){

        int[] myList = {9, 2, 3,5};
        // Print all the array elements
        for (int element: myList) {
            System.out.println(element); }
        }}
}
```

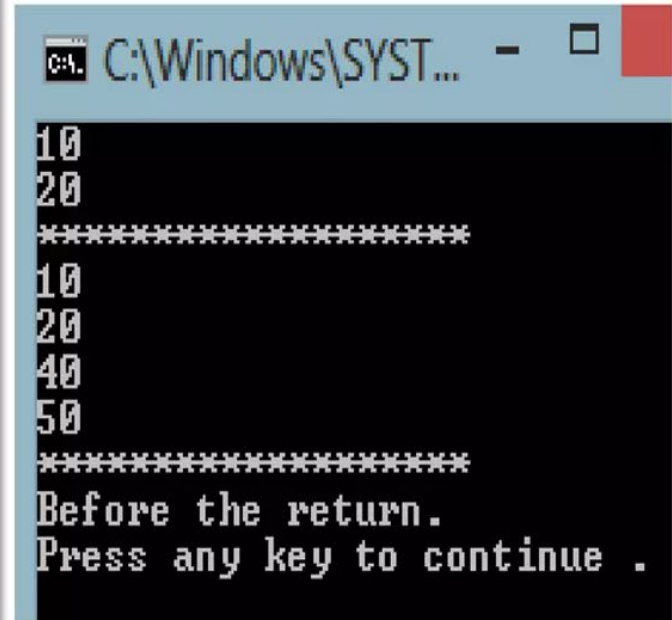


```
C:\Windows\SYSTEM... - □
9
2
3
5
Press any key to continue .
```

Jump in Array

- Java Supports 3 jump statements: break, continue and return.

```
public class Break_Array1d {  
    public static void main(String args[]) {  
        int[] numbers = {10, 20, 30, 40, 50};  
        for(int n : numbers )    {  
            if( n == 30 ) {  
                break;            }  
            System.out.println( n ); }  
        System.out.println("*****");  
        for(int n : numbers )    {  
            if( n == 30 ) {  
                continue;        }  
            System.out.println( n ); }  
        System.out.println("*****");  
        boolean t = true;  
        System.out.println("Before the return.");  
        if (t)  
            return;  
        //Compiler will bypass every statement after return  
        System.out.println("This won't execute.");  
    }  
}
```



```
C:\Windows\SYST...  
10  
20  
*****  
10  
20  
40  
50  
*****  
Before the return.  
Press any key to continue .
```

Example for String datatype and Sorting the array

```
import java.util.*;
public class Sort_Array_String {
public static void main(String args[]) {
String[] names ={"Sana", "Naz", "Lara", "Hama", "Shara"};
Arrays.sort(names);
for(String s : names )    {
System.out.println(s);
                        }

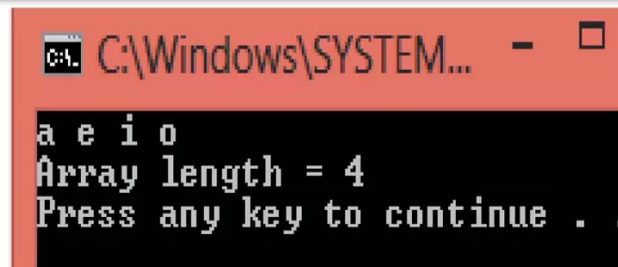
System.out.println("*****");
Arrays.sort(names, Collections.reverseOrder());
for(String s : names )    {
System.out.println(s);
                        }

}}
```

```
Hama
Lara
Naz
Sana
Shara
*****
Shara
Sana
Naz
Lara
Hama
Press any key to continue
```

Example of char datatype in an Array

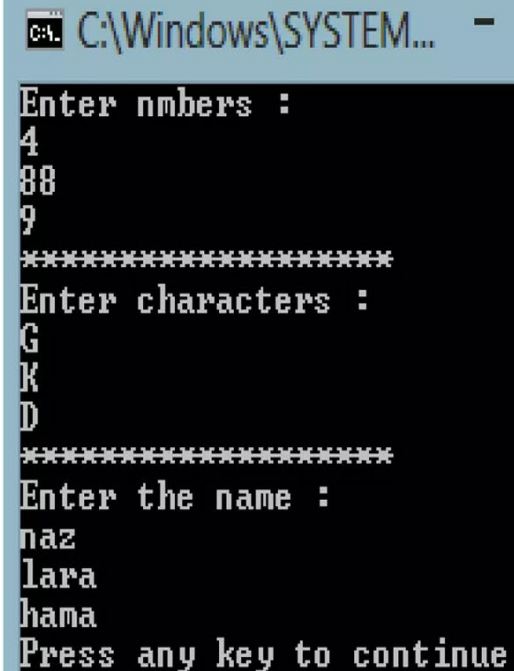
```
public class Char_Array{  
    public static void main(String args[]){  
        char[] letter = {'a', 'e', 'i', 'o'};  
        for(char ch : letter)  
        {  
            System.out.print(ch+" ");  
        }  
        System.out.println("");  
        int x = letter.length;  
        System.out.println("Array length = "+x);  
    }  
}
```



C:\Windows\SYSTEM...
a e i o
Array length = 4
Press any key to continue . .

User input in Array

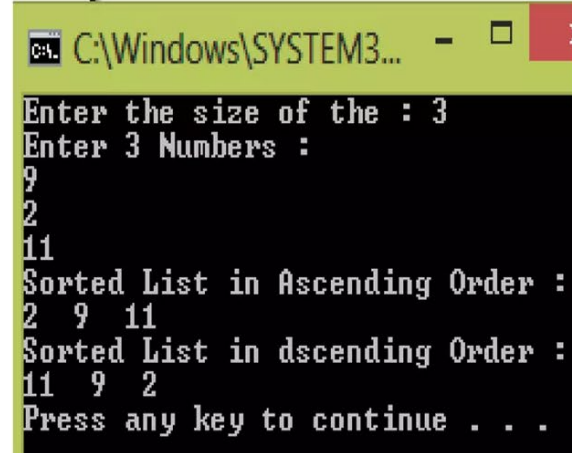
```
// Java program to read character using Scanner
import java.util.Scanner;
public class Array_Scanner{
public static void main(String[] args){
Scanner in = new Scanner(System.in);
int[] number = new int[3];
System.out.println("Enter numbers : ");
for (int i=0; i<3; i++) {
number[i] = in.nextInt(); }
System.out.println("*****");
char[] ch = new char[3];
System.out.println("Enter characters : ");
for (int i=0; i<3; i++) {
ch[i] = in.next().charAt(0);}
System.out.println("*****");
String[] names= new String[3];
System.out.println("Enter the name : ");
for (int ii=0;ii<3;ii++) {
names[ii] = in.next(); }
}}
```



```
C:\Windows\SYSTEM...
Enter numbers :
4
88
9
*****
Enter characters :
G
K
D
*****
Enter the name :
naz
lara
hama
Press any key to continue
```

1D Array Sorting in Ascending and Descending

```
import java.util.Scanner;
public class ArraySort_1d{
    public static void main(String args[]){
        Scanner scan = new Scanner(System.in);
        int n, i, j, temp;
        System.out.print("Enter the size of the : ");
        n = scan.nextInt();
        int arr[] = new int[n];
        System.out.println("Enter " +n+ " Numbers : ");
        for(i=0; i<n; i++) {
            arr[i] = scan.nextInt(); }
        //sorting the array Ascending
        for(i=0; i<(n-1); i++) {
            for(j=0; j<(n-i-1); j++) {
                if(arr[j] > arr[j+1]){
                    temp = arr[j];
                    arr[j] = arr[j+1];
                    arr[j+1] = temp; } } }
        System.out.print("Sorted List in Ascending Order : \n");
        for(i=0; i<n; i++) {
            System.out.print(arr[i]+ " ");
            System.out.println("");
        }
        //sorting the array descending
        System.out.print("Sorted List in dscending Order : \n");
        for(i=n-1; i>=0; i--) {
            System.out.print(arr[i]+ " ");
            System.out.println("");
        }
    }
}
```



```
C:\Windows\SYSTEM3...
Enter the size of the : 3
Enter 3 Numbers :
9
2
11
Sorted List in Ascending Order :
2 9 11
Sorted List in dscending Order :
11 9 2
Press any key to continue . . .
```


Multidimensional Array in Java

- Data is stored in row and column based index (Matrix Form)
- Syntax:
 - `dataType[][] arr;` or,
 - `dataType [][]arr;` or,
 - `dataType arr[][];` or,
 - `dataType[] arr[];`

Two dimensional array:

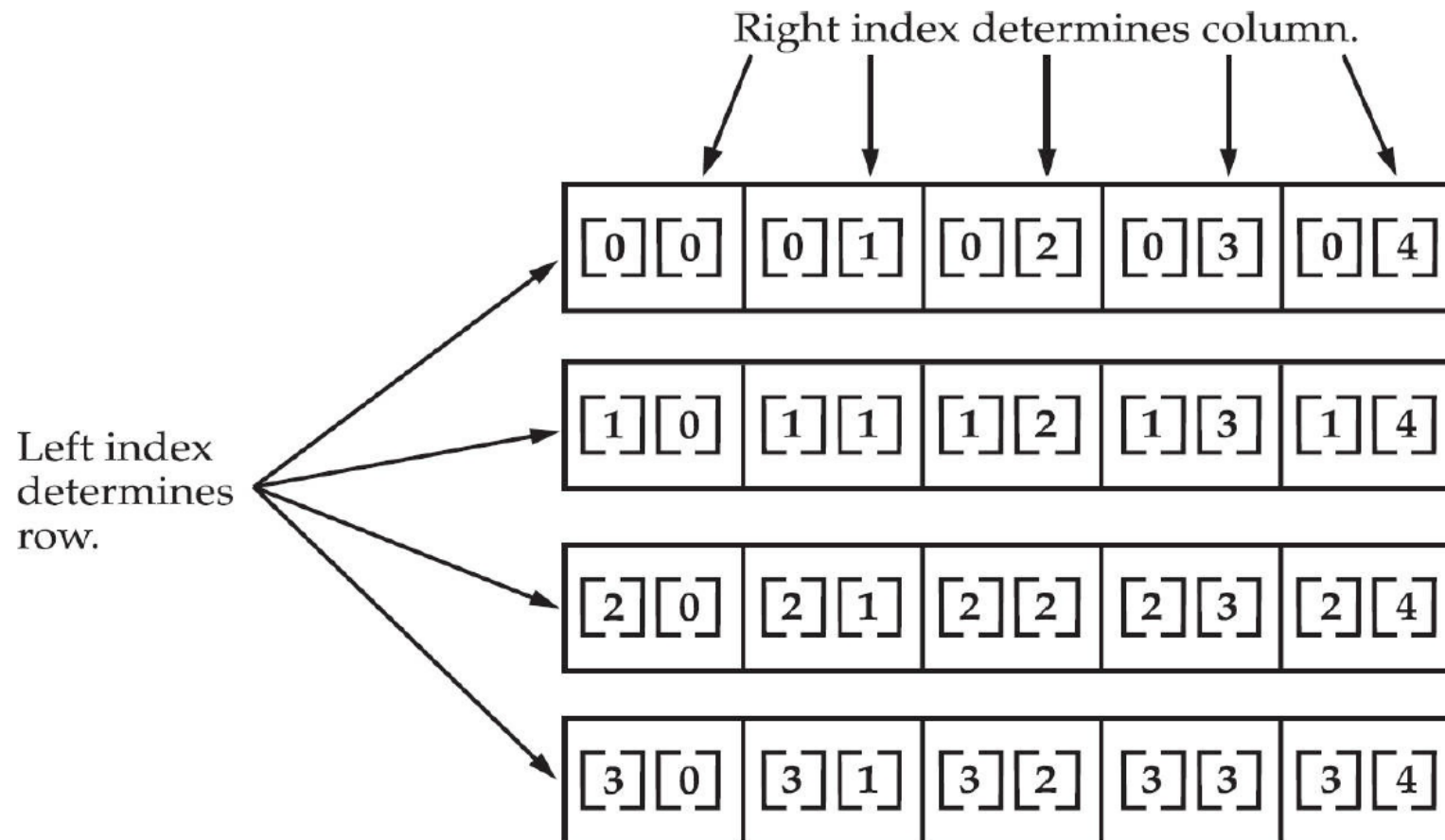
```
int[][] twoD_arr = new int[10][20];
```

Three dimensional array:

```
int[][][] threeD_arr = new int[10][20][30];
```

		Columns		
Rows	2D ARRAY	0	1	2
	0	a[0][0]	a[0][1]	a[0][2]
	1	a[1][0]	a[1][1]	a[1][2]
	2	a[2][0]	a[2][1]	a[2][2]

A conceptual view of a 4 by 5, two-dimensional array



Given: `int twoD [ ] [ ] = new int [4] [5] ;`

Multidimensional Array in Java Cont.

Example to instantiate Multidimensional Array in Java:

```
int[ ][ ] arr=new int[3][3]; //3 row and 3 column
```

Example to initialize Multidimensional Array in Java :

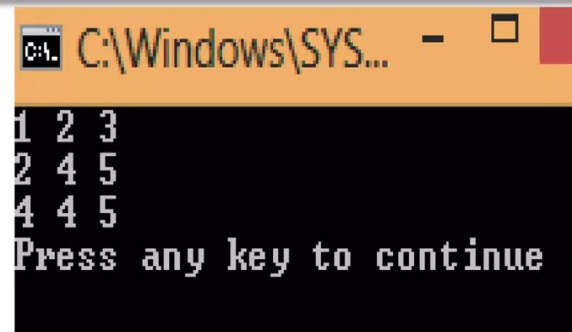
```
arr[0][0]=1; arr[0][1]=2; arr[0][2]=3;  
arr[1][0]=4; arr[1][1]=5; arr[1][2]=6;  
arr[2][0]=7; arr[2][1]=8; arr[2][2]=9;
```

Example of Multidimensional (2D) Array in Java

```
class Array_2D{
public static void main(String args[]){

int arr[][]={{1,2,3},{2,4,5},{4,4,5}};

for(int i=0;i<3;i++)      {
for(int j=0;j<3;j++)      {
    System.out.print(arr[i][j]+" "); }
    System.out.println();}
}}
```



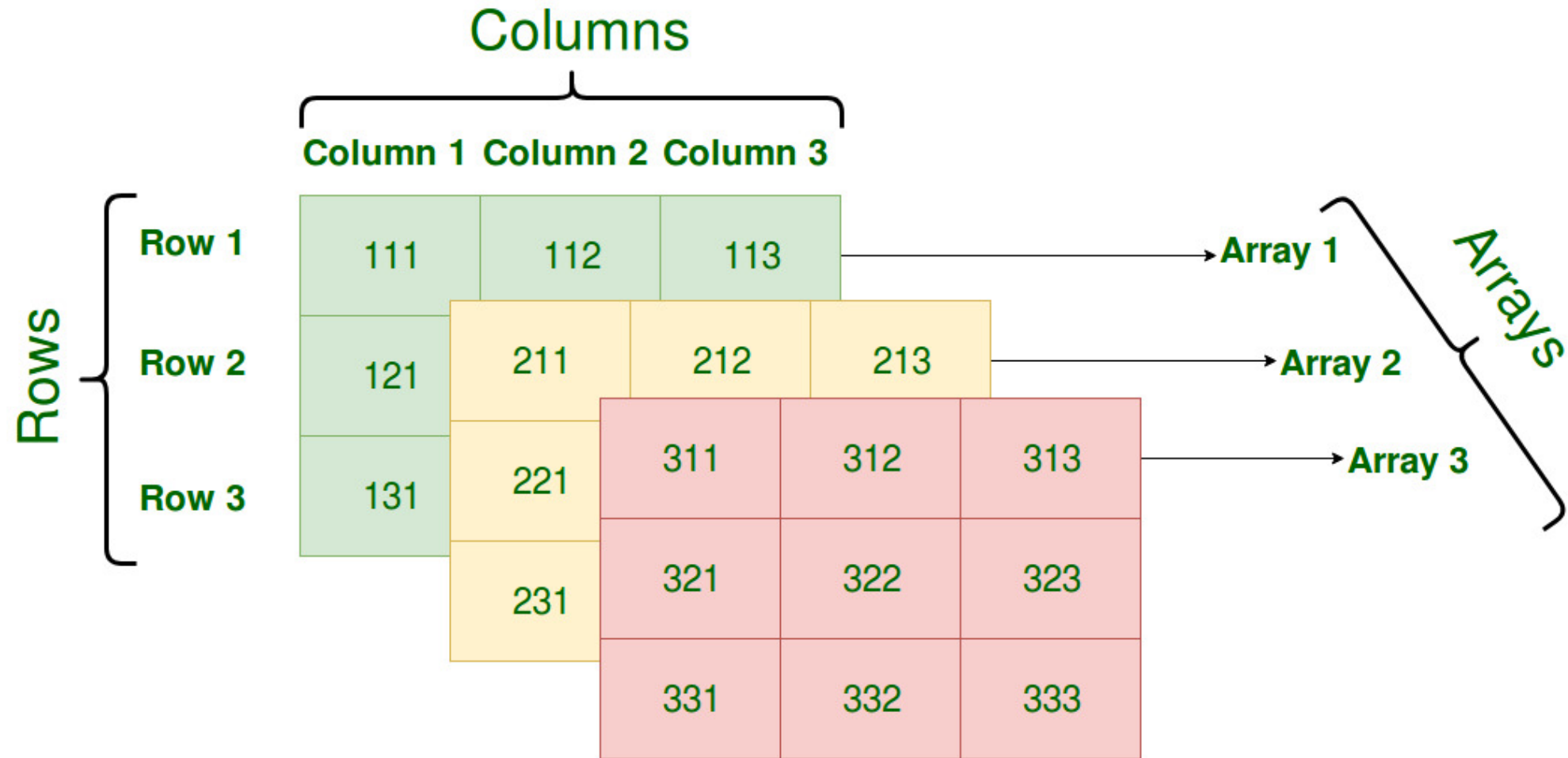
```
C:\Windows\SYS...
1 2 3
2 4 5
4 4 5
Press any key to continue .
```

Example of Multidimensional (2D) Array in Java using Scanner and Find Max number

```
import java.util.*;
class Array2d_Scanner{
public static void main(String[] args){
Scanner in=new Scanner(System.in);
int i,j,ii,jj;
int [][]a=new int[2][2];
for(i=0;i<a.length;i++) {
for(j=0;j<a.length;j++){
System.out.println("input value");
a[i][j]=in.nextInt(); } }
System.out.println("*****");
for(i=0;i<a.length;i++){
for(j=0;j<a.length;j++) {
System.out.print("\t"+a[i][j]+" ");}
System.out.println(); }
System.out.println("*****");
    int max=a[0][0];
    for(i=0;i<a.length;i++){
    for(j=0;j<a.length;j++){
    if(max<a[i][j])
    max=a[i][j]; }}
System.out.println("The max number= "+max);
}}
```

```
input value
4
input value
8
input value
11
input value
6
*****
           4           8
           11          6
*****
The max number= 11
Press any key to continue
```


Representation of 3D array in Tabular Format



Thank You